

EFFECT OF CRUDE EXTRACT OF *ALBIZIA PROCERA* ON GLYCEMIC CHANGE BEFORE AND AFTER GLUCOSE FEEDING IN NORMAL RABBITS

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ABSTRACT

The hypoglycemic effect of orally administered crude extract of *Albizia procera* leaves was studied in normoglycemic rabbits and in glucose loaded rabbits before and after glucose feeding. The extract administered at a dose 5 ml/kg was found to possess glucose lowering property in rabbits fed with glucose prior to the extract. In addition, to the plasma glucose, albumin, total lipids and total proteins were also determined.

INTRODUCTION

Plants have often been applied in traditional medicine as adjuvants in the therapy of diabetes mellitus. A large number of plant species alleged to exert hypoglycemic effects have been listed and described, for example, 18 Saudi plants (Al-Yahya *et al.*, 1984), 75 Indian plants (Nagarajan *et al.*, 1982), 7 European plants (Neef *et al.*, 1995), a comprehensive list of 342 plants (Atta-ur-Rahman and Zaman, 1989) and a brief description of 67 reputed plants (Handa *et al.*, 1989) and also a review of natural products as anti-diabetic drugs (Ivorra *et al.*, 1989).

The various plants parts of *Albizia procera* (family Mimosaceae) are used by traditional herbal medicinal practitioners in the subcontinent for the treatment of various diseases. For example, leaves are widely used as a home remedy for ulcer. The seeds are used for ophthalmia (Singh, 1970). It has been considered (with other ingredients) to be beneficial for stomach ache and for pregnant women. The moistened bark heals venomous stings of wasps and such (Perry and Metzger, 1980).

The leaves of *Albizia procera* locally known as "Sufaida" is one amongst the listed plants, without mentioning the nature of extract or active principle which possesses blood glucose lowering property, under medicinal plants of East and South Asia (Perry and Metzger, 1980; Atta-ur-Rahman and Zaman, 1989). This report deals with the effect of extract of *Albizia procera* leaves on the glycemic status of rabbits.

MATERIALS AND METHODS

Animals

Healthy rabbits of either sex (average weight 1.6-1.8 kg) purchased from the local market, kept in large and clean metallic cages (1-2 animals per cage) were fed a normal diet of lucerne and had free access to water for at least two weeks prior to experimentation.

Plant material

Fresh tender leaves of *Albizia procera* (350 g) were collected in the month of December of 1995 from the campus area of the University of Karachi. The leaves were soaked overnight in water and the next day were blended with water (1200 ml) in a kitchen blender. The extract was filtered and stored.

OGTT of control rabbits

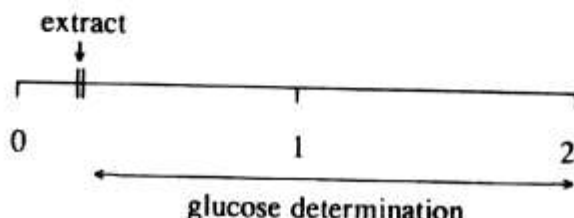
Oral glucose tolerance curve was established in normoglycemic rabbits by administering glucose (3g/kg body weight) and determining blood glucose at 0, 1 and 2 hours.

EXPERIMENTAL DESIGN

The animals were grouped as:

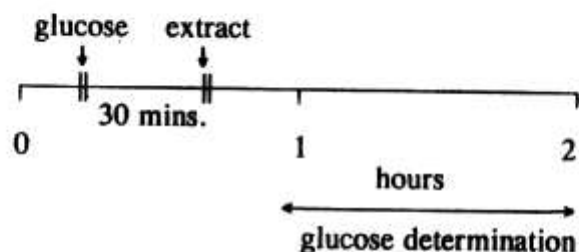
I. Controls: Controls for each group were administered physiological saline. A group of normal control that received equal amount of normal saline in place of glucose load/leaf extract was used and blood samples were collected according to the same schedule.

II. Extract treated: Overnight fasted rabbits were given an oral dose (5 ml/kg) of plant extract and plasma glucose concentration was determined at 0, 1 and 2 hours afterwards.

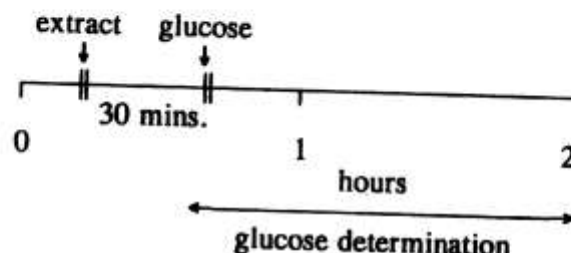


III. Glucose loaded rabbits

- (i) Overnight fasted rabbits were given an oral glucose load, 3 g/kg body weight in 10 ml. 30 minutes later, the crude extract (5 ml/kg) was orally administered. Plasma glucose concentration was determined at 0, 1 and 2 hours afterwards.



- (ii) Overnight fasting normoglycemic rabbits were orally given the crude extract (5 ml/kg) and 30 minutes later, an oral dose of glucose, 3 g/kg body weight (Al-Awadi *et al.*, 1985; Nayarko *et al.*, 1993; Sheela *et al.*, 1995) in 10 ml was given. Plasma glucose concentration was determined at 0, 1 and 2 hours afterwards.



Blood glucose determination

Blood (2 ml) was drawn out from the marginal ear veins of the rabbits into heparinized centrifuged tubes containing sodium fluoride (0.5 g), and glucose was determined by ortho-toluidine method (Dubowski *et al.*, 1970).

Glycemia change (%) was calculated as a time function as follows:

$$\% \text{ glycemia change} = \frac{G_x - G_o}{G_o} \times 100$$

G_o and G_x were the values of initial glycemia and glycemia at x minutes after treatment and the results were expressed as mg/100 ml.

RESULTS

OGTT of normoglycemic rabbits

The representative OGTT curves from some of the experiments were plotted. It was found that the mean glucose values (mg/dl) were 94.11 ± 14.61 (71.42-116.32) at 0, 132.71 ± 23.39 (93.87-151.02) at 1 and 101.97 ± 19.95 (79.06-134.80) at 2 hours following glucose ingestion. It was observed that there occurred 41% ingestion in glucose level at 1 hour which returned towards normal values at 2 hours.

Effect of crude extract

In two separate sets of experiments it was found that the crude extract decreased the level of glucose (mg/dl) from 108.50 ± 26.25 at 0 hr and 112.87 at 1 hr. to 84.44 ± 24.69 at 2 hr following the administration of the extract.

In order to understand the nature of hypoglycemic activity observed by simple administration two series of experiments were performed in different groups of animals in which the plant extract was given 30 minutes before or after glucose feeding.

(i) After glucose loading

In this group of normal rabbits, the mean plasma glucose (mg/dl) was 124.38 ± 23.81 at 0 hr. 97.67 ± 13.68 at 1 hr. and 85.26 ± 13.42 at 2 hr.

(ii) Prior to glucose loading

In rabbits loaded with glucose 30 minutes after extract administration, the mean plasma glucose (mg/dl) was 99.99 ± 24.83 at 0 hr. 145.34 ± 36.10 at 1 hr. and 89.91 ± 27.25 at 2 hr. A summary of the results of these experiments is presented in Fig. 1 and 2.

Albumin, protein and lipid content

Albumin (g/l), total lipids (mg/dl) and total proteins (g/dl) were determined in controls and was found to be 30.98 ± 5.64 , 200.16 ± 70.04 and 5.61 ± 1.09 respectively. Also, after 14 days of administration with *Albizia procera*, the protein content (g/dl) was 6.58 ± 0.34 which decreased to 4.64 ± 0.43 after 23 days of administration. Whereas, after 14 days of administration, the lipid content (mg/dl) was 166.66 ± 47.73 which decreased to 159.08 ± 45.45 after 23 days of administration. Thus, there was no significant change in the total lipids content after 14 and 23 days of administration of the extract.

Fig. 1: SHORT TERM EFFECT OF *A. PROCERA* IN NORMAL (I), GLUCOSE LOADED RABBITS AFTER (II) AND PRIOR TO (III) ADMINISTRATION OF GLUCOSE.

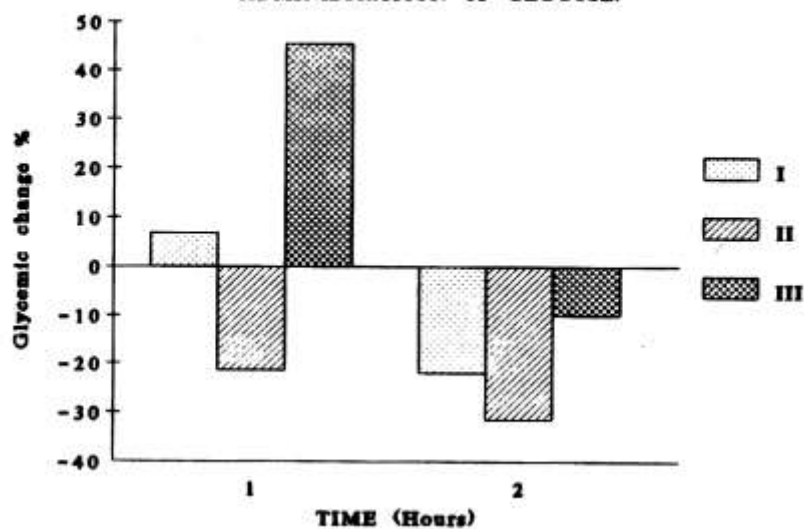
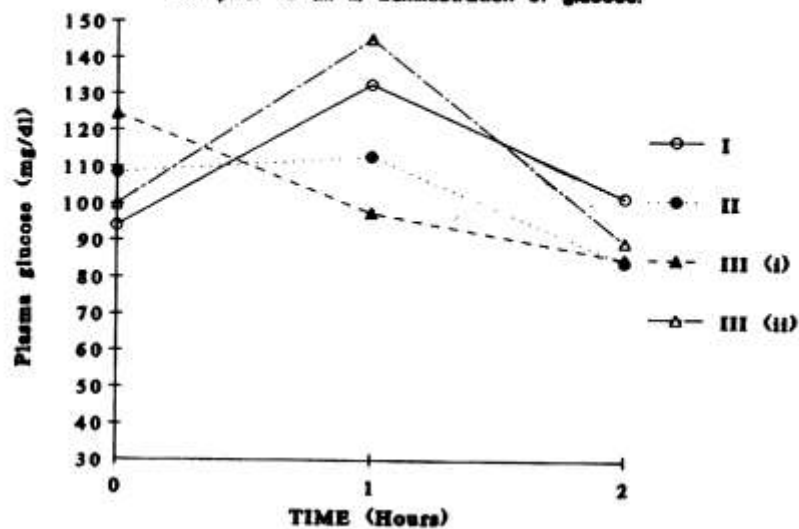


Fig. 2: Effect of *Abizia procera* on plasma glucose (mg/dl) in controls (I) Extract treated (II) and glucose loaded rabbits after (III I) and prior to (III II) administration of glucose.



DISCUSSION

The present study provides evidence that the plant *Albizia procera* exhibits hypoglycemic properties within 2 hr of their oral administration in both normal and glucose loaded rabbits.

For all the present experiments, the rabbits were acclimatized to room conditions prior to repeated experiments to establish optimized OGTT. These animals were found to exhibit normal OGTT curve with values of plasma glucose (mg/dl) as 94.11 ± 14.61 at 0 h, $132.71 \pm$ at 1 and 101.97 ± 19.95 at 2 hr; these were mostly in the range as established in many other labs (Jimenez *et al.*, 1991). Preliminary data shows that *Albizia procera* produced a marked decrease in plasma glucose of normoglycemic rabbits when its aqueous extract (5 ml/kg) was orally given. Although the effect was observed for a short duration, but repeated experiments have provided evidence that there was about 22% decrease in plasma glucose after 2 hr. of the oral administration of the extract. In order to understand the nature of glycemic activity, two series of experiments were performed in different groups of animals in which the plant extract was given 30 minutes before or after glucose load. The finding in the first group of normal animals which received the extract of *Albizia procera* orally administered 30 minutes after the glucose (3 g/kg body weight), there occurred a significant decrease after the 2 hr interval. It is clearly indicated that the level which was obtained at 1 hr after the extract given to glucose loaded rabbits was comparatively much lower than the controls i.e., animals which were given the crude extract only. Findings of 21.47% decrease at 1 hr which continued to 31.45% at 2 hr in these experiments provides support to the suggestion that this plant does exhibit glucose lowering property. Similar results were obtained previously for *swertia chirayita* (Baipai *et al.*, 1991) and *Gymnema sylvestre* (Chattopadhyay, 1993). In the reverse model (extract prior to glucose load), the finding of the plant *Albizia procera* showed a raised level of plasma glucose at 1 hr and decrease in the level at 2 hr interval, as compared to the fasting value at 0 hr. Thus, this experiment clearly prove that this plant possess glucose lowering components and hence should be placed amongst the hypoglycemic plants. Further, this experiment also demonstrates that it acts independent of glucose load. Similar properties were shown by the extracts of *Adiantum capillus veneris* (Neef *et al.*, 1995), *Atrocarpus heterophyllus* and *Asterocanthus longifolia* (Fernando *et al.*, 1991). The plant *Albizia procera* seems to fall in the category of hypoglycemic agents stimulating the pancreatic β -cells for insulin release.

In rabbits administered with *Albizia procera* it was observed that there was a marked decrease in the plasma protein content after 23 days as compared to the group which was administered the extract for 14 days. Similar results were also observed in animals which were administered with extracts of roots of *Salacia macroserma* (Venkatawelu *et al.*, 1993). The plasma lipid content

was also decreased after 23 days of administration with the extract of *Albizia procera*.

Thus, we have demonstrated that *Albizia procera* possesses a hypoglycemic effect which is evident in all experimental conditions, this suggests that the folk use of this plant, locally known as Sufaida, could be helpful in subjects suffering from wild form of diabetes mellitus, reducing the hyperglycemia produced by the intake of large amounts of carbohydrates, without producing any hypoglycemic effect in subjects having normal pancreatic function.

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